LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Demonstration of construction and explanation on operation of Synchro transmitters and receivers and Servo mechanisms.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Demonstration of construction and explanation on operation of Synchro transmitters and receivers and Servo mechanisms.
5. Specific Objectives	24.7.1.1 Trainee will be able to explain the operation of Synchro and Servo mechanisms.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Synchro Transmitter/Receiver wiring and Servo mechanism movement; Safety: Ensure power is off before connecting leads; Instructor supervision: Monitor trainee handling of delicate rotor/stator components.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a synchro system in aircraft control?
(b) How does a servo mechanism differ from a simple synchro connection?
(c) Why is precise angular position transmission critical for flight controls?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Synchro Transmitter Construction • Stator and Rotor Winding Configuration • AC Voltage Induction Principles • Signal Transmission to Receiver	L & Ds	PPT, IB	15 min
(b)	• Synchro Receiver Operation • Torque Motor Functionality • Error Signal Correction • Transmitter-Receiver Synchronization	L & Ds	PPT, PDF	20 min
(c)	• Servo Mechanism Components • Error Detection and Amplification • Feedback Loop Operation • Final Control Surface Movement	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the synchro transmitter converts angular position into an electrical signal.
(b) Describe the role of the torque motor in the synchro receiver system.
(c) How does the servo mechanism ensure accurate control surface positioning?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between a synchro system and a servo system, referencing the components discussed in this lesson to prepare for LP 09 on electrical measurements.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Measurement of various electrical and electronic parameters using an Oscilloscope.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min